

Field Data Review, Sale-level QA Summary

Mohawk Valley West Slope demonstration. 798 plots, 8722 trees, BAF 20 on a 300 ft grid across 24 units (1,621 gross ac, 1,408 net ac after stream exclusion zones). Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting. Standards are those of Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), which replaces the national sec. 41.1 figures for a Plumas sale. **Sale as a whole** (stratified by unit, area weights, $t = 2$): net cubic volume 7,086 cu ft/ac, sampling error 2.0 % at 95 % confidence (basal area 219 sq ft/ac, 1.1 %, reference only). R5 sec. 41.1 volume error standard 10.0 % for a tree-measurement sale valued at about \$1,685,031 on net acres (50,108 MBF at \$33.63/MBF, the Region 5 FY2025 average sold value, Forest Service Cut and Sold report; 57,649 MBF, \$1,938,608 on gross acres): **MEETS**. Basis: R5 Supplement 2409.12-2021-4 sec. 41.1, high-value sale of \$250,000 or more (10 % tree measurement, 20 % scaled); the national exhibit 01 tier for this value is 10 %. **Strata**: each unit's volume sampling error is tested against the R5 sec. 41.1 stratum standard of 30 % of volume for tree-measurement sales (40 % scaled; national text 41.1(5)(b): 40 %). The standard binds for any stratum over 10 % of total sale volume (Share %, Binds columns); smaller units are tested for information. 24 of 24 units meet it, worst 22.9 %; the standard binds for 0 unit(s). The basal-area error is listed for reference. **Measured trees** (R5 sec. 41.3): at least 20 measured trees per major species, major being 10 % or more of sale value (one stumpage rate, so value share equals net-acre volume share): PP 32.6 %, WF 30.4 %, DF 14.6 %, IC 12.6 %. Sale counts PP 2592, WF 2638, DF 1318, SP 846, IC 1325: **MEETS**; 23 of 24 units meet it on their own (R5 20 trees column). **Plots**: the R5 20-plot minimum for an area-based cruise (sec. 41.3); 20 of 24 units have it (R5 20-plot column). QA flags are plots that failed a record check: 7 flagged, 6 of 6 planted errors caught.

TABLE A. CRUISE STATISTICS AND QA STATUS BY UNIT (STRATUM)

Unit	Method	Acres	Plots	cu ft/ac	vol_cv	vol_se95	BA	ba_cv	ba_se95	t	Std %	Design	Share %	Binds	R5 20 trees	R5 20-plot	QA flags	Status
101	Tractor	97	45	6073	35	10.4	212	13	3.9	2.02	30	meets	4.4	info	ok	ok	7	follow-up
102	Tractor	36	18	6811	26	12.9	219	12	5.7	2.11	30	meets	1.6	info	ok	short	0	accepted
103	Tractor	53	27	6146	38	14.9	204	18	7.2	2.06	30	meets	3.0	info	ok	ok	0	accepted
401	Cable	30	14	7001	40	22.9	221	11	6.3	2.16	30	meets	1.8	info	short	short	0	accepted
104	Tractor	112	59	5141	37	9.6	207	15	3.9	2.00	30	meets	5.4	info	ok	ok	0	accepted
402	Cable	39	19	8094	30	14.4	214	14	6.6	2.10	30	meets	2.1	info	ok	short	0	accepted
105	Tractor	57	29	7800	23	8.7	222	11	4.3	2.05	30	meets	3.5	info	ok	ok	0	accepted
106	Tractor	89	41	7140	38	11.9	213	20	6.3	2.02	30	meets	5.3	info	ok	ok	0	accepted
107	Tractor	95	46	7915	23	6.8	227	11	3.3	2.01	30	meets	6.9	info	ok	ok	0	accepted
403	Cable	34	20	8263	18	8.2	229	11	5.2	2.09	30	meets	2.7	info	ok	ok	0	accepted
108	Tractor	63	31	7810	18	6.4	222	15	5.4	2.04	30	meets	4.5	info	ok	ok	0	accepted
109	Tractor	73	34	7627	28	9.7	214	25	8.8	2.03	30	meets	5.3	info	ok	ok	0	accepted
404	Cable	51	27	7814	30	12.0	208	19	7.6	2.06	30	meets	3.9	info	ok	ok	0	accepted
110	Tractor	62	31	5280	39	14.2	203	20	7.5	2.04	30	meets	3.0	info	ok	ok	0	accepted
111	Tractor	56	27	6509	34	13.3	222	18	7.1	2.06	30	meets	3.4	info	ok	ok	0	accepted
112	Tractor	80	40	6672	29	9.3	214	16	5.1	2.02	30	meets	4.1	info	ok	ok	0	accepted
113	Tractor	73	36	7904	18	6.1	228	12	4.1	2.03	30	meets	5.6	info	ok	ok	0	accepted
405	Cable	43	20	9406	27	12.6	236	10	4.7	2.09	30	meets	3.7	info	ok	ok	0	accepted
114	Tractor	116	56	6531	29	7.7	212	16	4.3	2.00	30	meets	6.2	info	ok	ok	0	accepted
115	Tractor	84	39	7132	25	8.1	228	12	4.0	2.02	30	meets	5.5	info	ok	ok	0	accepted
116	Tractor	89	44	8012	15	4.5	230	10	3.1	2.02	30	meets	6.7	info	ok	ok	0	accepted
117	Tractor	82	44	7022	25	7.7	223	13	4.0	2.02	30	meets	4.7	info	ok	ok	0	accepted
118	Tractor	70	34	7206	20	7.1	220	18	6.4	2.03	30	meets	4.8	info	ok	ok	0	accepted
406	Cable	35	17	8366	18	9.5	241	9	4.9	2.12	30	meets	1.9	info	ok	short	0	accepted

cu ft/ac: net cubic volume per acre (CVTS net of defect). vol_cv, ba_cv: coefficient of variation (%) of plot volume and plot basal area. vol_se95, ba_se95: sampling error (%), $t \times$ standard error / mean at 95 % confidence, Student's t on $n-1$ df. Std %: R5 stratum volume error standard. Design: vol_se95 against Std %. Share %: unit share of sale volume on net acres; Binds: yes where the share exceeds 10 %, info otherwise. R5 20 trees: the R5 20 measured trees per major species (by sale value), counted within the unit. R5 20-plot: the R5 20-plot minimum (sec. 41.3). BA in sq ft/ac. Sources: <https://www.fs.usda.gov/im/directives/field/r5/fsh/2409.12/R5-2409-12-40-2021-4.docx> ; <https://www.fs.usda.gov/im/directives/dughtml/fsh.html>

Unit 101 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 97.4 ac. 45 plots, BAF 20, variable-radius cruise on a 300 ft grid, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (1 SE of mean, BA 1.9 %)	Type	Sierran mixed conifer
Trees per acre	123	CWHR size / density	4 / D (cover 83 %)
QMD	17.8 in	SDI	288 (51 % of max 567)
SAWTIMBER (≥ 10 in)		BIOMASS (< 10 in)	
BA QMD	203 sq ft/ac 19.8 in	BA QMD	9 sq ft/ac 7.6 in
TPA	95	TPA	28
Volume	29.8 MBF/ac (5943 cu ft)	Volume	2.9 green tons/ac (130 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 146 sq ft/ac BA	Vol. sampling error	10.4 % at 95 % (CV 35 %, t = 2.02, n = 45)
Species (BA)	PP 70, WF 60, IC 32, DF 29, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 6
		Sale volume share	4.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.9 % at 95 % (CV 13 %); reference, not the standard
		Measured trees	PP 158, WF 134, DF 66, SP 46, IC 71; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	45; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	8	15.6	3.6	51	0.3
8-11.9	34	27.7	15.1	241	1.2
12-15.9	74	30.7	32.9	670	3.4
16-19.9	82	21.2	36.4	870	4.4
20-23.9	68	11.7	30.2	831	4.2
24-27.9	50	6.0	22.2	686	3.4
28-31.9	52	4.8	23.1	782	3.9
32-35.9	33	2.3	14.7	534	2.7
36-39.9	25	1.4	11.1	439	2.2
40-43.9	18	0.9	8.0	310	1.6
44-47.9	10	0.4	4.4	173	0.9
48-51.9	3	0.1	1.3	59	0.3
52-55.9	9	0.3	4.0	189	0.9
56-59.9	4	0.1	1.8	88	0.4
60-63.9	7	0.2	3.1	151	0.8
All	477	123.4	212.4	6073	30.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	158	45.2	70.2	16.9	2158	10.8
WF	134	31.5	59.6	18.6	1701	8.5
IC	71	19.1	31.6	17.4	768	3.9
DF	66	14.3	29.3	19.4	849	4.3
SP	47	13.3	20.9	17.0	574	2.9
PPP	1	0.1	0.4	35.1	23	0.1
All	477	123.4	212.4	17.8	6073	30.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101 plots, 300 ft grid (orange: QA flag)

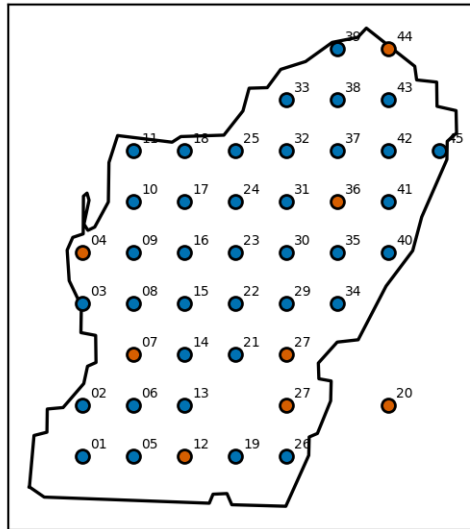


Figure 1. Plot locations with QA status.

QA findings: 7 item(s) for crew follow-up

Plot	Finding
101-04	Tree 1: species code 'PPP' not valid. Correct in edit.
101-07	Tree 9: height 32 ft outside tolerance for DBH 27.8. Remeasure.
101-12	Tree 1: DBH 999.0 out of range. Remeasure.
101-20	Plot falls outside the unit boundary. Check GPS position and unit number.
101-27	Plot id recorded 2 times. Check plot numbers.
101-36	Tree 2: height missing. Measure.
101-44	Tree 1: height 15 ft outside tolerance for DBH 34.0. Remeasure.

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
101-01	12	240	11	ok
101-02	10	200	14	ok
101-03	6	120	28	ok
101-04	9	180	25	FLAG
101-05	9	180	16	ok
101-06	12	240	14	ok
101-07	13	260	9	FLAG
101-08	11	220	10	ok
101-09	11	220	19	ok
101-10	12	240	25	ok
101-11	9	180	24	ok
101-12	10	200	8	FLAG
101-13	11	220	5	ok
101-14	10	200	7	ok
101-15	10	200	3	ok
101-16	11	220	6	ok
101-17	11	220	4	ok
101-18	10	200	10	ok
101-19	10	200	8	ok
101-20	11	220	9	FLAG
101-21	11	220	10	ok
101-22	10	200	5	ok
101-23	11	220	3	ok
101-24	11	220	5	ok
101-25	11	220	6	ok
101-26	15	300	15	ok
101-27	11	220	9	FLAG
101-27	9	180	17	FLAG
101-29	12	240	8	ok
101-30	11	220	3	ok
101-31	11	220	3	ok
101-32	11	220	4	ok
101-33	9	180	7	ok
101-34	9	180	9	ok
101-35	10	200	2	ok
101-36	12	240	3	FLAG
101-37	11	220	6	ok
101-38	11	220	14	ok
101-39	10	200	4	ok
101-40	12	240	1	ok
101-41	12	240	2	ok
101-42	10	200	4	ok
101-43	9	180	8	ok
101-44	11	220	6	FLAG
101-45	10	200	10	ok

Unit 102 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 36.3 ac. 18 plots, BAF 20, variable-radius cruise on a 300 ft grid, 197 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	219 sq ft/ac (1 SE of mean, BA 2.7 %)	Type	Sierran mixed conifer
Trees per acre	88	CWHR size / density	4 / D (cover 87 %)
QMD	21.3 in	SDI	285 (50 % of max 574)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	218 sq ft/ac 21.5 in	BA QMD	1 sq ft/ac 9.8 in
TPA	86	TPA	2
Volume	34.1 MBF/ac (6796 cu ft)	Volume	0.3 green tons/ac (15 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 155 sq ft/ac BA	Vol. sampling error	12.9 % at 95 % (CV 26 %, t = 2.11, n = 18)
Species (BA)	PP 69, WF 66, DF 33, IC 28, SP 23	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	1.6 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	5.7 % at 95 % (CV 12 %); reference, not the standard
		Measured trees	PP 62, WF 59, DF 30, SP 21, IC 25; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	18; R5 20-plot minimum for an area-based cruise (sec. 41.3): SHORT by 2

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	10.7	6.7	124	0.6
12-15.9	18	19.6	20.0	394	2.0
16-19.9	32	20.2	35.6	897	4.5
20-23.9	40	17.1	44.4	1299	6.5
24-27.9	22	6.7	24.4	770	3.9
28-31.9	30	7.1	33.3	1104	5.5
32-35.9	19	3.4	21.1	744	3.7
36-39.9	8	1.1	8.9	365	1.8
40-43.9	10	1.2	11.1	441	2.2
44-47.9	5	0.5	5.6	240	1.2
48-51.9	1	0.1	1.1	66	0.3
52-55.9	3	0.2	3.3	180	0.9
56-59.9	1	0.1	1.1	73	0.4
60-63.9	2	0.1	2.2	115	0.6
All	197	88.1	218.9	6811	34.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	62	28.1	68.9	21.2	2266	11.4
WF	59	26.4	65.6	21.3	2042	10.2
DF	30	15.3	33.3	20.0	1016	5.1
IC	25	9.1	27.8	23.6	755	3.8
SP	21	9.2	23.3	21.6	733	3.7
All	197	88.1	218.9	21.3	6811	34.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 102 plots, 300 ft grid (orange: QA flag)

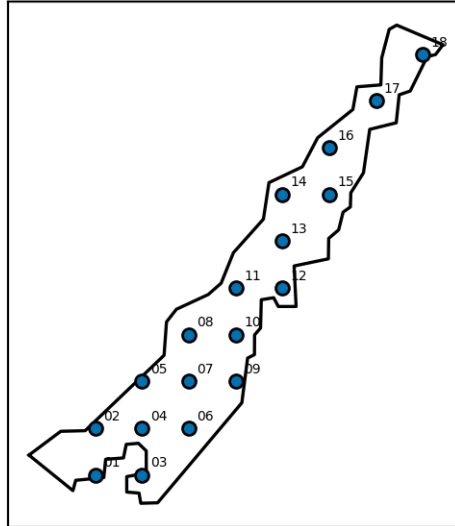


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted; 2 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
102-01	12	240	4	ok
102-02	11	220	19	ok
102-03	9	180	5	ok
102-04	11	220	5	ok
102-05	10	200	15	ok
102-06	12	240	4	ok
102-07	8	160	4	ok
102-08	12	240	27	ok
102-09	10	200	5	ok
102-10	11	220	8	ok
102-11	12	240	19	ok
102-12	12	240	9	ok
102-13	10	200	6	ok
102-14	11	220	19	ok
102-15	10	200	4	ok
102-16	12	240	22	ok
102-17	11	220	19	ok
102-18	13	260	36	ok

Unit 103 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 53.4 ac. 27 plots, BAF 20, variable-radius cruise on a 300 ft grid, 276 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	204 sq ft/ac (1 SE of mean, BA 3.5 %)	Type	Sierran mixed conifer
Trees per acre	108	CWHR size / density	4 / D (cover 80 %)
QMD	18.6 in	SDI	273 (47 % of max 585)
SAWTIMBER (≥ 10 in)		BIOMASS (< 10 in)	
BA QMD	196 sq ft/ac 20.6 in	BA QMD	8 sq ft/ac 8.0 in
TPA	85	TPA	23
Volume	30.3 MBF/ac (6028 cu ft)	Volume	2.7 green tons/ac (118 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 153 sq ft/ac BA	Vol. sampling error	14.9 % at 95 % (CV 38 %, t = 2.06, n = 27)
Species (BA)	WF 65, PP 58, DF 37, IC 27, SP 18	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 7
		Sale volume share	3.0 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	7.2 % at 95 % (CV 18 %); reference, not the standard
		Measured trees	PP 78, WF 88, DF 50, SP 24, IC 36; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	27; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	4	10.6	3.0	44	0.2
8-11.9	19	26.0	14.1	216	1.1
12-15.9	36	24.8	26.7	581	2.9
16-19.9	41	17.7	30.4	727	3.6
20-23.9	46	13.0	34.1	1001	5.0
24-27.9	32	6.5	23.7	739	3.7
28-31.9	26	4.0	19.3	591	3.0
32-35.9	17	2.0	12.6	497	2.5
36-39.9	15	1.4	11.1	460	2.3
40-43.9	12	1.0	8.9	335	1.7
44-47.9	8	0.5	5.9	270	1.4
48-51.9	10	0.5	7.4	319	1.6
52-55.9	2	0.1	1.5	80	0.4
56-59.9	3	0.1	2.2	103	0.5
60-63.9	5	0.2	3.7	182	0.9
All	276	108.4	204.4	6146	30.9

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	88	33.8	65.2	18.8	2087	10.5
PP	78	33.6	57.8	17.8	1818	9.1
DF	50	18.3	37.0	19.2	1076	5.4
IC	36	13.6	26.7	19.0	663	3.3
SP	24	9.1	17.8	18.9	502	2.5
All	276	108.4	204.4	18.6	6146	30.9

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 103 plots, 300 ft grid (orange: QA flag)

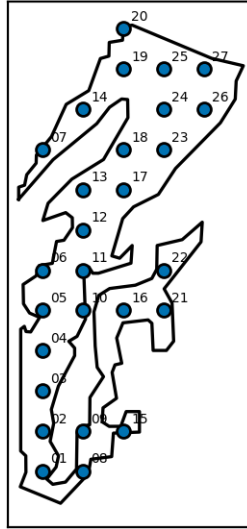


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
103-01	9	180	25	ok
103-02	11	220	19	ok
103-03	14	280	32	ok
103-04	10	200	33	ok
103-05	12	240	36	ok
103-06	8	160	28	ok
103-07	12	240	33	ok
103-08	10	200	22	ok
103-09	7	140	32	ok
103-10	9	180	29	ok
103-11	11	220	37	ok
103-12	8	160	26	ok
103-13	10	200	25	ok
103-14	13	260	30	ok
103-15	11	220	27	ok
103-16	9	180	22	ok
103-17	10	200	23	ok
103-18	12	240	25	ok
103-19	9	180	22	ok
103-20	7	140	16	ok
103-21	7	140	19	ok
103-22	10	200	35	ok
103-23	11	220	18	ok
103-24	12	240	16	ok
103-25	12	240	21	ok
103-26	12	240	14	ok
103-27	10	200	5	ok

Unit 401 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 29.5 ac. 14 plots, BAF 20, variable-radius cruise on a 300 ft grid, 155 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	221 sq ft/ac (1 SE of mean, BA 2.9 %)	Type	Sierran mixed conifer
Trees per acre	101	CWHR size / density	4 / D (cover 94 %)
QMD	20.0 in	SDI	290 (49 % of max 591)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	217 sq ft/ac 20.8 in	BA QMD	4 sq ft/ac 9.5 in
TPA	92	TPA	9
Volume	34.8 MBF/ac (6933 cu ft)	Volume	1.5 green tons/ac (68 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 158 sq ft/ac BA	Vol. sampling error	22.9 % at 95 % (CV 40 %, t = 2.16, n = 14)
Species (BA)	WF 79, PP 66, DF 33, IC 24, SP 20	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 9
		Sale volume share	1.8 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	6.3 % at 95 % (CV 11 %); reference, not the standard
		Measured trees	PP 46, WF 55, DF 23, SP 14, IC 17; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): SHORT : IC 17
		Plot count	14; R5 20-plot minimum for an area-based cruise (sec. 41.3): SHORT by 6

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	9	21.0	12.9	194	1.0
12-15.9	27	35.9	38.6	755	3.8
16-19.9	19	15.2	27.1	654	3.3
20-23.9	15	8.4	21.4	552	2.8
24-27.9	19	7.6	27.1	927	4.7
28-31.9	14	4.1	20.0	669	3.4
32-35.9	23	5.2	32.9	1269	6.4
36-39.9	8	1.5	11.4	512	2.6
40-43.9	1	0.1	1.4	50	0.3
44-47.9	5	0.6	7.1	320	1.6
48-51.9	10	1.1	14.3	725	3.6
52-55.9	2	0.2	2.9	178	0.9
56-59.9	1	0.1	1.4	61	0.3
60-63.9	2	0.1	2.9	137	0.7
All	155	101.1	221.4	7001	35.1

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	55	31.7	78.6	21.3	2481	12.5
PP	46	30.6	65.7	19.8	2347	11.8
DF	23	13.8	32.9	20.9	1043	5.2
IC	17	12.9	24.3	18.6	542	2.7
SP	14	12.0	20.0	17.5	588	3.0
All	155	101.1	221.4	20.0	7001	35.1

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

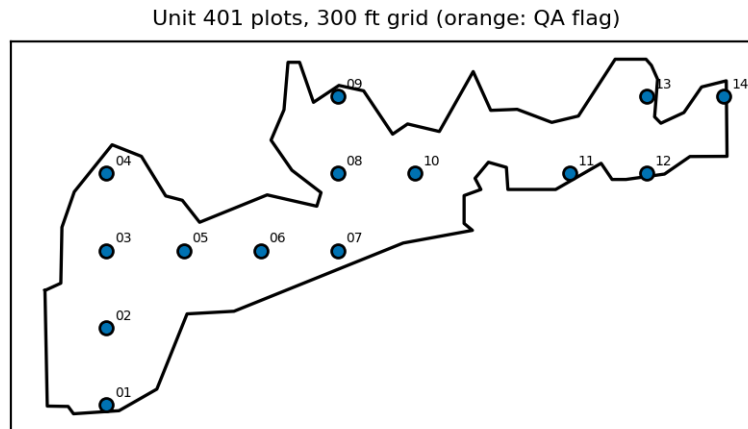


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted; under the R5 20 measured trees per major species for IC (sec. 41.3); 6 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
401-01	12	240	51	ok
401-02	10	200	46	ok
401-03	11	220	47	ok
401-04	13	260	37	ok
401-05	12	240	40	ok
401-06	10	200	38	ok
401-07	11	220	43	ok
401-08	13	260	36	ok
401-09	10	200	38	ok
401-10	11	220	38	ok
401-11	11	220	36	ok
401-12	9	180	49	ok
401-13	10	200	38	ok
401-14	12	240	46	ok

Unit 104 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 111.5 ac. 59 plots, BAF 20, variable-radius cruise on a 300 ft grid, 611 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	207 sq ft/ac (1 SE of mean, BA 2.0 %)	Type	Sierran mixed conifer
Trees per acre	165	CWHR size / density	4 / D (cover 85 %)
QMD	15.2 in	SDI	300 (52 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	186 sq ft/ac 18.2 in	BA QMD	21 sq ft/ac 7.8 in
TPA	103	TPA	62
Volume	24.2 MBF/ac (4817 cu ft)	Volume	7.4 green tons/ac (324 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 138 sq ft/ac BA	Vol. sampling error	9.6 % at 95 % (CV 37 %, t = 2.00, n = 59)
Species (BA)	PP 66, WF 61, IC 32, DF 29, SP 20	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 7
		Sale volume share	5.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.9 % at 95 % (CV 15 %); reference, not the standard
		Measured trees	PP 195, WF 179, DF 86, SP 58, IC 93; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	59; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	21	31.5	7.1	116	0.6
8-11.9	79	50.4	26.8	411	2.1
12-15.9	108	34.9	36.6	682	3.4
16-19.9	121	23.9	41.0	943	4.7
20-23.9	81	10.5	27.5	712	3.6
24-27.9	78	7.4	26.4	758	3.8
28-31.9	38	2.7	12.9	427	2.1
32-35.9	30	1.6	10.2	364	1.8
36-39.9	24	1.1	8.1	305	1.5
40-43.9	12	0.4	4.1	161	0.8
44-47.9	7	0.2	2.4	99	0.5
48-51.9	6	0.2	2.0	82	0.4
52-55.9	2	0.0	0.7	26	0.1
56-59.9	2	0.0	0.7	27	0.1
60-63.9	2	0.0	0.7	29	0.1
All	611	164.8	207.1	5141	25.8

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	195	49.2	66.1	15.7	1799	9.0
WF	179	51.6	60.7	14.7	1527	7.7
IC	93	24.7	31.5	15.3	654	3.3
DF	86	20.6	29.2	16.1	702	3.5
SP	58	18.5	19.7	14.0	459	2.3
All	611	164.8	207.1	15.2	5141	25.8

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 104 plots, 300 ft grid (orange: QA flag)

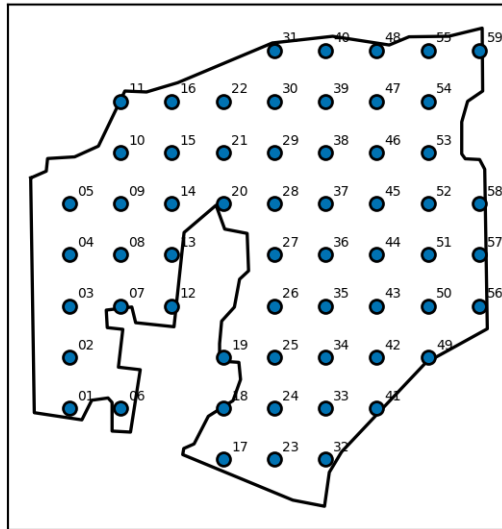


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-01	8	160	20	ok
104-02	12	240	10	ok
104-03	12	240	7	ok
104-04	11	220	9	ok
104-05	6	120	7	ok
104-06	10	200	24	ok
104-07	8	160	15	ok
104-08	11	220	9	ok
104-09	8	160	5	ok
104-10	9	180	9	ok
104-11	9	180	30	ok
104-12	10	200	33	ok
104-13	11	220	21	ok
104-14	10	200	10	ok
104-15	9	180	5	ok
104-16	8	160	10	ok
104-17	11	220	15	ok
104-18	12	240	20	ok
104-19	9	180	17	ok
104-20	12	240	33	ok
104-21	9	180	9	ok
104-22	11	220	8	ok
104-23	11	220	24	ok
104-24	10	200	18	ok
104-25	12	240	27	ok
104-26	10	200	32	ok
104-27	13	260	21	ok
104-28	11	220	13	ok
104-29	8	160	12	ok
104-30	11	220	6	ok
104-31	11	220	22	ok
104-32	11	220	13	ok
104-33	11	220	11	ok
104-34	8	160	14	ok
104-35	12	240	15	ok
104-36	13	260	13	ok
104-37	11	220	35	ok
104-38	11	220	17	ok
104-39	10	200	7	ok
104-40	11	220	10	ok
104-41	11	220	22	ok
104-42	12	240	9	ok
104-43	10	200	13	ok
104-44	12	240	6	ok
104-45	11	220	13	ok
104-46	9	180	34	ok
104-47	12	240	7	ok
104-48	9	180	22	ok
104-49	10	200	29	ok
104-50	6	120	11	ok
104-51	12	240	5	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-52	12	240	4	ok
104-53	11	220	16	ok
104-54	10	200	11	ok
104-55	12	240	4	ok
104-56	11	220	13	ok
104-57	11	220	5	ok
104-58	10	200	12	ok
104-59	9	180	2	ok

SIMULATED DATA

Unit 402 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 38.9 ac. 19 plots, BAF 20, variable-radius cruise on a 300 ft grid, 203 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 3.1 %)	Type	Sierran mixed conifer
Trees per acre	64	CWHR size / density	5 / D (cover 90 %)
QMD	24.8 in	SDI	261 (45 % of max 576)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	214 sq ft/ac 24.8 in	BA QMD	0 sq ft/ac 0.0 in
TPA	64	TPA	0
Volume	40.6 MBF/ac (8094 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 165 sq ft/ac BA	Vol. sampling error	14.4 % at 95 % (CV 30 %, t = 2.10, n = 19)
Species (BA)	PP 67, WF 65, DF 36, IC 27, SP 18	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 5
		Sale volume share	2.1 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	6.6 % at 95 % (CV 14 %); reference, not the standard
		Measured trees	PP 64, WF 62, DF 34, SP 17, IC 26; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	19; R5 20-plot minimum for an area-based cruise (sec. 41.3): SHORT by 1

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
12-15.9	14	13.1	14.7	360	1.8
16-19.9	30	17.5	31.6	872	4.4
20-23.9	33	13.0	34.7	1056	5.3
24-27.9	28	8.0	29.5	1017	5.1
28-31.9	17	3.6	17.9	659	3.3
32-35.9	12	2.0	12.6	534	2.7
36-39.9	20	2.7	21.1	900	4.5
40-43.9	10	1.1	10.5	500	2.5
44-47.9	15	1.4	15.8	811	4.1
48-51.9	5	0.4	5.3	269	1.3
52-55.9	3	0.2	3.2	171	0.9
56-59.9	4	0.2	4.2	241	1.2
60-63.9	12	0.6	12.6	706	3.5
All	203	63.8	213.7	8094	40.6

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	64	16.8	67.4	27.2	2968	14.9
WF	62	20.7	65.3	24.1	2377	11.9
DF	34	12.0	35.8	23.4	1254	6.3
IC	26	9.5	27.4	23.0	818	4.1
SP	17	4.8	17.9	26.1	677	3.4
All	203	63.8	213.7	24.8	8094	40.6

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 402 plots, 300 ft grid (orange: QA flag)

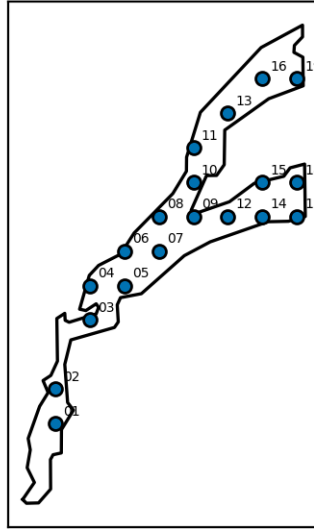


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted; 1 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
402-01	7	140	37	ok
402-02	12	240	53	ok
402-03	10	200	44	ok
402-04	12	240	61	ok
402-05	12	240	83	ok
402-06	9	180	36	ok
402-07	11	220	48	ok
402-08	10	200	56	ok
402-09	10	200	34	ok
402-10	12	240	58	ok
402-11	9	180	39	ok
402-12	11	220	76	ok
402-13	10	200	61	ok
402-14	11	220	62	ok
402-15	10	200	42	ok
402-16	12	240	57	ok
402-17	12	240	46	ok
402-18	13	260	51	ok
402-19	10	200	57	ok

Unit 105 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 57.4 ac. 29 plots, BAF 20, variable-radius cruise on a 300 ft grid, 322 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (1 SE of mean, BA 2.1 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	274 (48 % of max 565)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	221 sq ft/ac 24.8 in	BA QMD	1 sq ft/ac 7.2 in
TPA	66	TPA	5
Volume	39.0 MBF/ac (7775 cu ft)	Volume	0.6 green tons/ac (25 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 161 sq ft/ac BA	Vol. sampling error	8.7 % at 95 % (CV 23 %, t = 2.05, n = 29)
Species (BA)	PP 72, WF 60, DF 34, IC 32, SP 23	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 3
		Sale volume share	3.5 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.3 % at 95 % (CV 11 %); reference, not the standard
		Measured trees	PP 105, WF 87, DF 50, SP 34, IC 46; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	29; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	3.2	0.7	12	0.1
8-11.9	5	6.1	3.4	65	0.3
12-15.9	16	9.6	11.0	266	1.3
16-19.9	36	13.9	24.8	675	3.4
20-23.9	50	13.2	34.5	1039	5.2
24-27.9	46	8.5	31.7	1022	5.1
28-31.9	48	6.8	33.1	1174	5.9
32-35.9	43	4.8	29.7	1078	5.4
36-39.9	18	1.6	12.4	521	2.6
40-43.9	17	1.2	11.7	475	2.4
44-47.9	10	0.6	6.9	312	1.6
48-51.9	11	0.6	7.6	381	1.9
52-55.9	4	0.2	2.8	151	0.8
56-59.9	6	0.2	4.1	215	1.1
60-63.9	11	0.4	7.6	415	2.1
All	322	70.9	222.1	7800	39.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	105	21.5	72.4	24.9	2883	14.5
WF	87	22.1	60.0	22.3	2011	10.1
DF	50	12.0	34.5	23.0	1114	5.6
IC	46	7.6	31.7	27.6	928	4.7
SP	34	7.7	23.4	23.6	864	4.3
All	322	70.9	222.1	24.0	7800	39.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 105 plots, 300 ft grid (orange: QA flag)

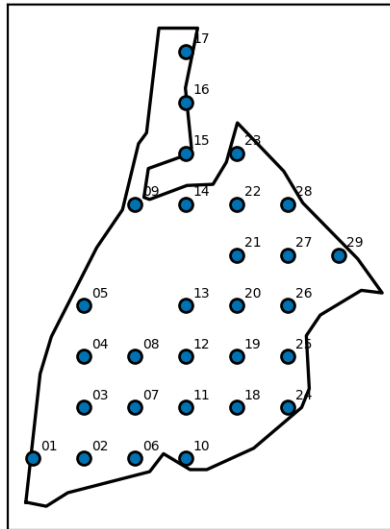


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
105-01	11	220	11	ok
105-02	12	240	18	ok
105-03	12	240	14	ok
105-04	14	280	32	ok
105-05	10	200	20	ok
105-06	10	200	16	ok
105-07	11	220	6	ok
105-08	12	240	26	ok
105-09	11	220	29	ok
105-10	13	260	22	ok
105-11	11	220	20	ok
105-12	10	200	20	ok
105-13	10	200	22	ok
105-14	10	200	19	ok
105-15	9	180	33	ok
105-16	10	200	36	ok
105-17	11	220	28	ok
105-18	11	220	12	ok
105-19	8	160	22	ok
105-20	12	240	28	ok
105-21	11	220	26	ok
105-22	12	240	18	ok
105-23	13	260	26	ok
105-24	12	240	15	ok
105-25	10	200	18	ok
105-26	11	220	28	ok
105-27	12	240	13	ok
105-28	12	240	19	ok
105-29	11	220	16	ok

Unit 106 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 88.6 ac. 41 plots, BAF 20, variable-radius cruise on a 300 ft grid, 437 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	213 sq ft/ac (1 SE of mean, BA 3.1 %)	Type	Sierran mixed conifer
Trees per acre	95	CWHR size / density	4 / D (cover 85 %)
QMD	20.3 in	SDI	272 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	207 sq ft/ac 22.9 in	BA QMD	6 sq ft/ac 7.0 in
TPA	73	TPA	22
Volume	35.5 MBF/ac (7064 cu ft)	Volume	1.7 green tons/ac (77 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 159 sq ft/ac BA	Vol. sampling error	11.9 % at 95 % (CV 38 %, t = 2.02, n = 41)
Species (BA)	WF 68, PP 65, IC 32, DF 29, SP 19	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 7
		Sale volume share	5.3 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	6.3 % at 95 % (CV 20 %); reference, not the standard
		Measured trees	PP 134, WF 139, DF 59, SP 39, IC 66; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	41; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	9	18.8	4.4	56	0.3
8-11.9	12	10.2	5.9	96	0.5
12-15.9	39	18.5	19.0	427	2.1
16-19.9	55	15.2	26.8	714	3.6
20-23.9	49	9.3	23.9	682	3.4
24-27.9	64	8.6	31.2	1027	5.2
28-31.9	53	5.3	25.9	918	4.6
32-35.9	51	4.0	24.9	934	4.7
36-39.9	28	1.8	13.7	567	2.8
40-43.9	29	1.5	14.1	581	2.9
44-47.9	11	0.5	5.4	249	1.2
48-51.9	15	0.5	7.3	377	1.9
52-55.9	6	0.2	2.9	126	0.6
56-59.9	5	0.1	2.4	103	0.5
60-63.9	11	0.3	5.4	283	1.4
All	437	94.7	213.2	7140	35.8

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	139	29.2	67.8	20.6	2279	11.4
PP	134	27.8	65.4	20.7	2465	12.4
IC	66	11.9	32.2	22.2	891	4.5
DF	59	13.9	28.8	19.5	898	4.5
SP	39	11.8	19.0	17.2	608	3.0
All	437	94.7	213.2	20.3	7140	35.8

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 106 plots, 300 ft grid (orange: QA flag)

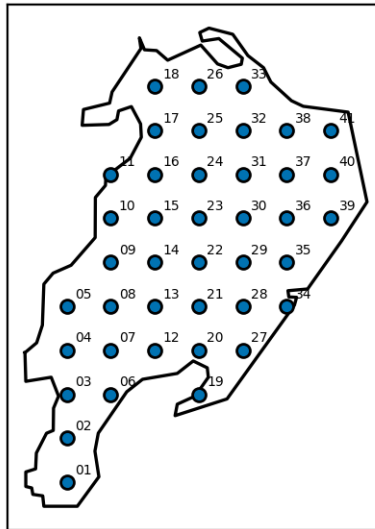


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
106-01	9	180	14	ok
106-02	7	140	9	ok
106-03	12	240	14	ok
106-04	13	260	10	ok
106-05	13	260	9	ok
106-06	11	220	12	ok
106-07	12	240	21	ok
106-08	8	160	6	ok
106-09	10	200	5	ok
106-10	14	280	11	ok
106-11	13	260	11	ok
106-12	8	160	33	ok
106-13	7	140	10	ok
106-14	10	200	11	ok
106-15	13	260	11	ok
106-16	11	220	8	ok
106-17	9	180	21	ok
106-18	11	220	19	ok
106-19	12	240	34	ok
106-20	12	240	28	ok
106-21	7	140	29	ok
106-22	12	240	18	ok
106-23	12	240	5	ok
106-24	11	220	6	ok
106-25	10	200	7	ok
106-26	13	260	18	ok
106-27	11	220	28	ok
106-28	10	200	28	ok
106-29	10	200	12	ok
106-30	10	200	12	ok
106-31	11	220	10	ok
106-32	11	220	2	ok
106-33	12	240	14	ok
106-34	11	220	28	ok
106-35	12	240	10	ok
106-36	12	240	9	ok
106-37	11	220	7	ok
106-38	11	220	8	ok
106-39	3	60	4	ok
106-40	10	200	10	ok
106-41	12	240	8	ok

Unit 107 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 95.3 ac. 46 plots, BAF 20, variable-radius cruise on a 300 ft grid, 522 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	227 sq ft/ac (1 SE of mean, BA 1.6 %)	Type	Sierran mixed conifer
Trees per acre	72	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	280 (48 % of max 584)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	226 sq ft/ac 24.4 in	BA QMD	1 sq ft/ac 8.9 in
TPA	69	TPA	3
Volume	39.6 MBF/ac (7889 cu ft)	Volume	0.6 green tons/ac (25 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 165 sq ft/ac BA	Vol. sampling error	6.8 % at 95 % (CV 23 %, t = 2.01, n = 46)
Species (BA)	WF 71, PP 65, IC 37, DF 33, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 3
		Sale volume share	6.9 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.3 % at 95 % (CV 11 %); reference, not the standard
		Measured trees	PP 149, WF 164, DF 76, SP 48, IC 85; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	46; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	5.6	3.0	63	0.3
12-15.9	33	13.0	14.3	327	1.6
16-19.9	62	15.5	27.0	722	3.6
20-23.9	79	13.5	34.3	1049	5.3
24-27.9	82	9.9	35.7	1203	6.0
28-31.9	61	5.5	26.5	914	4.6
32-35.9	54	3.8	23.5	873	4.4
36-39.9	42	2.4	18.3	724	3.6
40-43.9	30	1.4	13.0	566	2.8
44-47.9	18	0.7	7.8	344	1.7
48-51.9	18	0.6	7.8	345	1.7
52-55.9	14	0.4	6.1	311	1.6
56-59.9	8	0.2	3.5	153	0.8
60-63.9	14	0.3	6.1	321	1.6
All	522	72.5	227.0	7915	39.7

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	164	22.5	71.3	24.1	2504	12.6
PP	149	20.1	64.8	24.3	2487	12.5
IC	85	13.1	37.0	22.8	1064	5.3
DF	76	10.2	33.0	24.4	1079	5.4
SP	48	6.5	20.9	24.2	780	3.9
All	522	72.5	227.0	24.0	7915	39.7

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 107 plots, 300 ft grid (orange: QA flag)

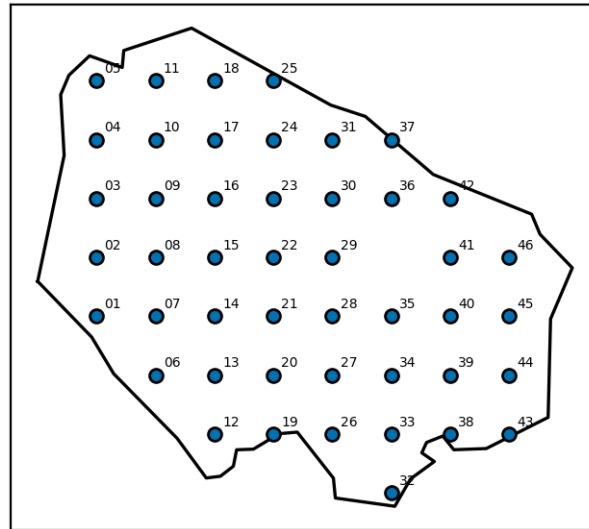


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
107-01	13	260	11	ok
107-02	12	240	5	ok
107-03	13	260	31	ok
107-04	11	220	12	ok
107-05	8	160	20	ok
107-06	11	220	6	ok
107-07	11	220	9	ok
107-08	13	260	6	ok
107-09	12	240	21	ok
107-10	10	200	17	ok
107-11	12	240	18	ok
107-12	11	220	16	ok
107-13	10	200	20	ok
107-14	13	260	39	ok
107-15	13	260	19	ok
107-16	11	220	15	ok
107-17	9	180	23	ok
107-18	12	240	9	ok
107-19	12	240	25	ok
107-20	10	200	29	ok
107-21	12	240	22	ok
107-22	12	240	23	ok
107-23	11	220	29	ok
107-24	12	240	9	ok
107-25	10	200	27	ok
107-26	8	160	28	ok
107-27	10	200	15	ok
107-28	11	220	26	ok
107-29	12	240	36	ok
107-30	11	220	8	ok
107-31	11	220	16	ok
107-32	12	240	25	ok
107-33	10	200	29	ok
107-34	11	220	12	ok
107-35	11	220	17	ok
107-36	11	220	14	ok
107-37	12	240	22	ok
107-38	12	240	34	ok
107-39	12	240	30	ok
107-40	12	240	20	ok
107-41	12	240	13	ok
107-42	13	260	30	ok
107-43	14	280	33	ok
107-44	10	200	23	ok
107-45	11	220	18	ok
107-46	12	240	16	ok

Unit 403 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 34.0 ac. 20 plots, BAF 20, variable-radius cruise on a 300 ft grid, 229 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	229 sq ft/ac (1 SE of mean, BA 2.5 %)	Type	Sierran mixed conifer
Trees per acre	63	CWHR size / density	5 / D (cover 95 %)
QMD	25.9 in	SDI	277 (49 % of max 562)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	229 sq ft/ac 25.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	63	TPA	0
Volume	41.5 MBF/ac (8263 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 163 sq ft/ac BA	Vol. sampling error	8.2 % at 95 % (CV 18 %, t = 2.09, n = 20)
Species (BA)	PP 68, WF 53, DF 40, IC 38, SP 30	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	2.7 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	5.2 % at 95 % (CV 11 %); reference, not the standard
		Measured trees	PP 68, WF 53, DF 40, SP 30, IC 38; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	20; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.5	1.0	20	0.1
12-15.9	9	8.4	9.0	236	1.2
16-19.9	23	13.2	23.0	652	3.3
20-23.9	34	12.9	34.0	1052	5.3
24-27.9	33	8.9	33.0	1143	5.7
28-31.9	29	6.0	29.0	981	4.9
32-35.9	28	4.4	28.0	1048	5.3
36-39.9	25	3.2	25.0	989	5.0
40-43.9	18	1.9	18.0	775	3.9
44-47.9	12	1.1	12.0	557	2.8
48-51.9	4	0.3	4.0	176	0.9
52-55.9	4	0.3	4.0	180	0.9
56-59.9	5	0.3	5.0	253	1.3
60-63.9	4	0.2	4.0	201	1.0
All	229	62.6	229.0	8263	41.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	68	19.1	68.0	25.5	2804	14.1
WF	53	14.9	53.0	25.5	1895	9.5
DF	40	10.0	40.0	27.1	1379	6.9
IC	38	9.2	38.0	27.6	1125	5.6
SP	30	9.4	30.0	24.2	1059	5.3
All	229	62.6	229.0	25.9	8263	41.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

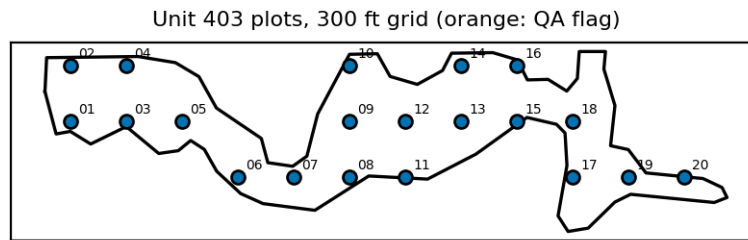


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
403-01	12	240	42	ok
403-02	13	260	69	ok
403-03	13	260	37	ok
403-04	13	260	48	ok
403-05	9	180	36	ok
403-06	10	200	38	ok
403-07	10	200	40	ok
403-08	10	200	43	ok
403-09	12	240	40	ok
403-10	13	260	39	ok
403-11	12	240	37	ok
403-12	12	240	36	ok
403-13	11	220	42	ok
403-14	11	220	42	ok
403-15	10	200	36	ok
403-16	12	240	37	ok
403-17	13	260	48	ok
403-18	11	220	47	ok
403-19	10	200	46	ok
403-20	12	240	35	ok

Unit 108 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 63.3 ac. 31 plots, BAF 20, variable-radius cruise on a 300 ft grid, 344 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (1 SE of mean, BA 2.6 %)	Type	Sierran mixed conifer
Trees per acre	68	CWHR size / density	5 / D (cover 91 %)
QMD	24.5 in	SDI	274 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	222 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	68	TPA	0
Volume	39.2 MBF/ac (7810 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 164 sq ft/ac BA	Vol. sampling error	6.4 % at 95 % (CV 18 %, t = 2.04, n = 31)
Species (BA)	WF 68, PP 66, IC 36, DF 29, SP 22	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	4.5 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	5.4 % at 95 % (CV 15 %); reference, not the standard
		Measured trees	PP 103, WF 106, DF 45, SP 34, IC 56; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	31; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	3.6	2.6	66	0.3
12-15.9	22	13.2	14.2	345	1.7
16-19.9	33	12.0	21.3	610	3.1
20-23.9	53	12.7	34.2	1063	5.3
24-27.9	60	10.7	38.7	1307	6.6
28-31.9	44	5.9	28.4	1021	5.1
32-35.9	39	4.0	25.2	919	4.6
36-39.9	35	2.9	22.6	896	4.5
40-43.9	26	1.7	16.8	701	3.5
44-47.9	6	0.3	3.9	167	0.8
48-51.9	7	0.3	4.5	212	1.1
52-55.9	6	0.2	3.9	209	1.1
56-59.9	5	0.2	3.2	153	0.8
60-63.9	4	0.1	2.6	140	0.7
All	344	68.1	221.9	7810	39.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	106	22.5	68.4	23.6	2344	11.8
PP	103	19.0	66.5	25.3	2641	13.3
IC	56	12.9	36.1	22.7	993	5.0
DF	45	7.0	29.0	27.6	1002	5.0
SP	34	6.7	21.9	24.4	830	4.2
All	344	68.1	221.9	24.5	7810	39.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 108 plots, 300 ft grid (orange: QA flag)

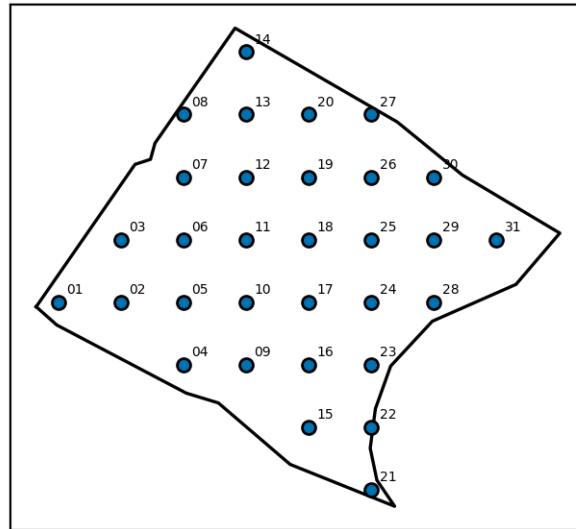


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
108-01	10	200	11	ok
108-02	11	220	22	ok
108-03	9	180	23	ok
108-04	10	200	18	ok
108-05	10	200	16	ok
108-06	12	240	21	ok
108-07	11	220	27	ok
108-08	11	220	23	ok
108-09	11	220	27	ok
108-10	12	240	12	ok
108-11	11	220	14	ok
108-12	14	280	25	ok
108-13	6	120	17	ok
108-14	13	260	6	ok
108-15	14	280	15	ok
108-16	10	200	13	ok
108-17	11	220	9	ok
108-18	11	220	8	ok
108-19	9	180	32	ok
108-20	12	240	17	ok
108-21	10	200	22	ok
108-22	12	240	30	ok
108-23	11	220	18	ok
108-24	12	240	11	ok
108-25	10	200	16	ok
108-26	13	260	15	ok
108-27	13	260	19	ok
108-28	11	220	10	ok
108-29	11	220	13	ok
108-30	10	200	20	ok
108-31	13	260	10	ok

Unit 109 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 72.9 ac. 34 plots, BAF 20, variable-radius cruise on a 300 ft grid, 364 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 4.3 %)	Type	Sierran mixed conifer
Trees per acre	65	CWHR size / density	5 / D (cover 88 %)
QMD	24.6 in	SDI	263 (44 % of max 592)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	213 sq ft/ac 25.1 in	BA QMD	1 sq ft/ac 8.8 in
TPA	62	TPA	3
Volume	38.1 MBF/ac (7598 cu ft)	Volume	0.6 green tons/ac (28 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 169 sq ft/ac BA	Vol. sampling error	9.7 % at 95 % (CV 28 %, t = 2.03, n = 34)
Species (BA)	WF 70, PP 55, DF 36, IC 32, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	5.3 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	8.8 % at 95 % (CV 25 %); reference, not the standard
		Measured trees	PP 94, WF 119, DF 62, SP 35, IC 54; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	34; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	4.3	2.4	61	0.3
12-15.9	17	9.4	10.0	233	1.2
16-19.9	40	12.7	23.5	661	3.3
20-23.9	68	15.6	40.0	1236	6.2
24-27.9	44	7.1	25.9	830	4.2
28-31.9	49	6.0	28.8	1076	5.4
32-35.9	39	3.7	22.9	875	4.4
36-39.9	31	2.3	18.2	748	3.8
40-43.9	25	1.5	14.7	631	3.2
44-47.9	13	0.7	7.6	325	1.6
48-51.9	9	0.4	5.3	234	1.2
52-55.9	9	0.3	5.3	250	1.3
56-59.9	6	0.2	3.5	169	0.9
60-63.9	10	0.3	5.9	298	1.5
All	364	64.7	214.1	7627	38.3

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	119	21.1	70.0	24.7	2480	12.5
PP	94	16.0	55.3	25.2	2184	11.0
DF	62	11.2	36.5	24.4	1245	6.3
IC	54	10.5	31.8	23.5	929	4.7
SP	35	5.9	20.6	25.3	789	4.0
All	364	64.7	214.1	24.6	7627	38.3

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 109 plots, 300 ft grid (orange: QA flag)

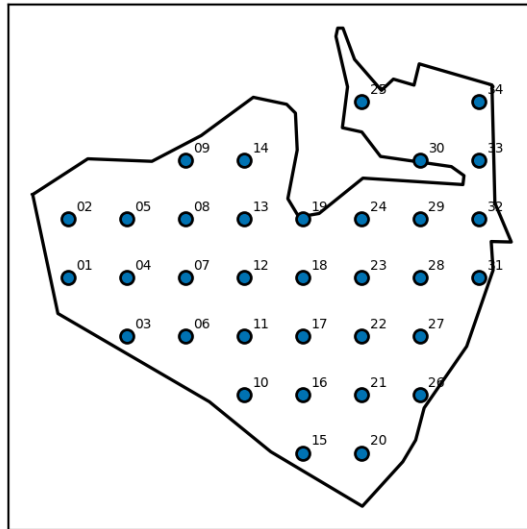


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
109-01	15	300	8	ok
109-02	10	200	7	ok
109-03	10	200	21	ok
109-04	11	220	7	ok
109-05	9	180	7	ok
109-06	12	240	19	ok
109-07	3	60	15	ok
109-08	12	240	4	ok
109-09	12	240	17	ok
109-10	11	220	9	ok
109-11	13	260	16	ok
109-12	11	220	6	ok
109-13	15	300	11	ok
109-14	13	260	3	ok
109-15	10	200	17	ok
109-16	10	200	7	ok
109-17	10	200	11	ok
109-18	10	200	7	ok
109-19	7	140	35	ok
109-20	12	240	16	ok
109-21	12	240	8	ok
109-22	9	180	7	ok
109-23	11	220	5	ok
109-24	13	260	24	ok
109-25	10	200	24	ok
109-26	10	200	21	ok
109-27	10	200	11	ok
109-28	14	280	12	ok
109-29	11	220	21	ok
109-30	11	220	35	ok
109-31	13	260	25	ok
109-32	2	40	21	ok
109-33	12	240	34	ok
109-34	10	200	14	ok

Unit 404 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 51.2 ac. 27 plots, BAF 20, variable-radius cruise on a 300 ft grid, 281 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	208 sq ft/ac (1 SE of mean, BA 3.7 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 87 %)
QMD	25.0 in	SDI	252 (43 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	207 sq ft/ac 25.7 in	BA QMD	1 sq ft/ac 5.9 in
TPA	57	TPA	4
Volume	39.2 MBF/ac (7803 cu ft)	Volume	0.2 green tons/ac (10 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 168 sq ft/ac BA	Vol. sampling error	12.0 % at 95 % (CV 30 %, t = 2.06, n = 27)
Species (BA)	WF 67, PP 64, DF 30, IC 27, SP 20	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 5
		Sale volume share	3.9 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	7.6 % at 95 % (CV 19 %); reference, not the standard
		Measured trees	PP 87, WF 90, DF 41, SP 27, IC 36; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	27; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	3.9	0.7	10	0.1
8-11.9	4	4.4	3.0	50	0.3
12-15.9	10	6.5	7.4	181	0.9
16-19.9	27	11.5	20.0	555	2.8
20-23.9	36	10.2	26.7	802	4.0
24-27.9	43	8.9	31.9	1062	5.3
28-31.9	38	5.8	28.1	1080	5.4
32-35.9	37	4.3	27.4	1059	5.3
36-39.9	22	2.1	16.3	699	3.5
40-43.9	13	1.0	9.6	411	2.1
44-47.9	17	1.1	12.6	599	3.0
48-51.9	17	0.9	12.6	654	3.3
52-55.9	3	0.1	2.2	115	0.6
56-59.9	3	0.1	2.2	110	0.6
60-63.9	10	0.4	7.4	425	2.1
All	281	61.3	208.1	7814	39.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	90	23.7	66.7	22.7	2425	12.2
PP	87	16.6	64.4	26.6	2704	13.6
DF	41	7.5	30.4	27.2	1099	5.5
IC	36	6.7	26.7	27.0	792	4.0
SP	27	6.7	20.0	23.4	793	4.0
All	281	61.3	208.1	25.0	7814	39.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 404 plots, 300 ft grid (orange: QA flag)

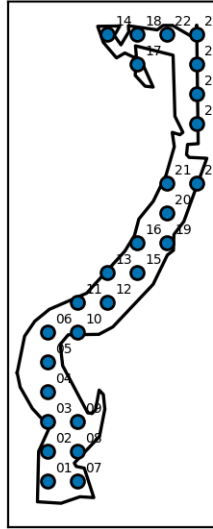


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
404-01	11	220	65	ok
404-02	11	220	50	ok
404-03	13	260	39	ok
404-04	10	200	67	ok
404-05	11	220	57	ok
404-06	11	220	56	ok
404-07	12	240	54	ok
404-08	9	180	48	ok
404-09	13	260	60	ok
404-10	6	120	41	ok
404-11	12	240	52	ok
404-12	11	220	60	ok
404-13	12	240	44	ok
404-14	9	180	48	ok
404-15	8	160	59	ok
404-16	10	200	58	ok
404-17	11	220	36	ok
404-18	10	200	47	ok
404-19	4	80	46	ok
404-20	10	200	71	ok
404-21	11	220	57	ok
404-22	10	200	43	ok
404-23	10	200	40	ok
404-24	10	200	57	ok
404-25	11	220	50	ok
404-26	12	240	50	ok
404-27	13	260	36	ok

Unit 110 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 62.0 ac. 31 plots, BAF 20, variable-radius cruise on a 300 ft grid, 315 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	203 sq ft/ac (1 SE of mean, BA 3.7 %)	Type	Sierran mixed conifer
Trees per acre	137	CWHR size / density	4 / D (cover 82 %)
QMD	16.5 in	SDI	287 (50 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	194 sq ft/ac 18.4 in	BA QMD	10 sq ft/ac 7.5 in
TPA	105	TPA	32
Volume	25.8 MBF/ac (5138 cu ft)	Volume	3.2 green tons/ac (142 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 142 sq ft/ac BA	Vol. sampling error	14.2 % at 95 % (CV 39 %, t = 2.04, n = 31)
Species (BA)	PP 65, WF 60, IC 28, DF 26, SP 25	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 7
		Sale volume share	3.0 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	7.5 % at 95 % (CV 20 %); reference, not the standard
		Measured trees	PP 101, WF 93, DF 40, SP 38, IC 43; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	31; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	6	18.9	3.9	56	0.3
8-11.9	27	31.5	17.4	290	1.5
12-15.9	60	36.7	38.7	766	3.8
16-19.9	68	25.4	43.9	1030	5.2
20-23.9	46	11.5	29.7	732	3.7
24-27.9	30	5.4	19.4	584	2.9
28-31.9	32	4.3	20.6	646	3.2
32-35.9	15	1.6	9.7	319	1.6
36-39.9	10	0.8	6.5	243	1.2
40-43.9	5	0.3	3.2	135	0.7
44-47.9	5	0.3	3.2	145	0.7
48-51.9	3	0.1	1.9	86	0.4
52-55.9	3	0.1	1.9	85	0.4
56-59.9	2	0.1	1.3	60	0.3
60-63.9	3	0.1	1.9	103	0.5
All	315	137.0	203.2	5280	26.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	101	46.5	65.2	16.0	1774	8.9
WF	93	36.5	60.0	17.4	1570	7.9
IC	43	21.4	27.7	15.4	615	3.1
DF	40	16.1	25.8	17.1	693	3.5
SP	38	16.5	24.5	16.5	628	3.2
All	315	137.0	203.2	16.5	5280	26.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 110 plots, 300 ft grid (orange: QA flag)

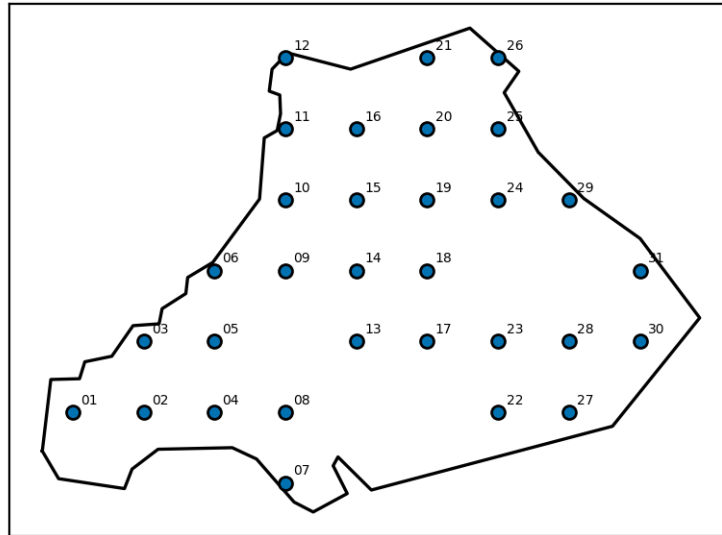


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
110-01	6	120	25	ok
110-02	13	260	21	ok
110-03	11	220	17	ok
110-04	8	160	19	ok
110-05	7	140	27	ok
110-06	10	200	20	ok
110-07	13	260	35	ok
110-08	13	260	30	ok
110-09	12	240	17	ok
110-10	6	120	27	ok
110-11	10	200	19	ok
110-12	7	140	23	ok
110-13	9	180	31	ok
110-14	11	220	34	ok
110-15	10	200	23	ok
110-16	11	220	32	ok
110-17	12	240	16	ok
110-18	10	200	35	ok
110-19	10	200	14	ok
110-20	12	240	27	ok
110-21	13	260	14	ok
110-22	11	220	8	ok
110-23	11	220	15	ok
110-24	10	200	27	ok
110-25	10	200	17	ok
110-26	8	160	29	ok
110-27	12	240	33	ok
110-28	10	200	21	ok
110-29	12	240	25	ok
110-30	7	140	35	ok
110-31	10	200	28	ok

Unit 111 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 55.7 ac. 27 plots, BAF 20, variable-radius cruise on a 300 ft grid, 300 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (1 SE of mean, BA 3.5 %)	Type	Sierran mixed conifer
Trees per acre	110	CWHR size / density	4 / D (cover 86 %)
QMD	19.2 in	SDI	296 (51 % of max 586)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	216 sq ft/ac 20.9 in	BA QMD	7 sq ft/ac 7.9 in
TPA	91	TPA	20
Volume	32.2 MBF/ac (6422 cu ft)	Volume	2.0 green tons/ac (87 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 154 sq ft/ac BA	Vol. sampling error	13.3 % at 95 % (CV 34 %, t = 2.06, n = 27)
Species (BA)	WF 71, PP 63, DF 36, IC 35, SP 18	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 6
		Sale volume share	3.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	7.1 % at 95 % (CV 18 %); reference, not the standard
		Measured trees	PP 85, WF 96, DF 48, SP 24, IC 47; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	27; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	12.6	3.7	50	0.3
8-11.9	13	17.1	9.6	145	0.7
12-15.9	36	25.6	26.7	561	2.8
16-19.9	46	19.8	34.1	866	4.3
20-23.9	49	14.1	36.3	1027	5.2
24-27.9	43	8.7	31.9	965	4.8
28-31.9	38	5.9	28.1	933	4.7
32-35.9	25	3.0	18.5	642	3.2
36-39.9	16	1.5	11.9	489	2.5
40-43.9	11	0.9	8.1	302	1.5
44-47.9	13	0.8	9.6	355	1.8
48-51.9	2	0.1	1.5	59	0.3
52-55.9	2	0.1	1.5	69	0.3
60-63.9	1	0.0	0.7	44	0.2
All	300	110.2	222.2	6509	32.7

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	96	34.7	71.1	19.4	2117	10.6
PP	85	32.9	63.0	18.7	1993	10.0
DF	48	16.5	35.6	19.9	977	4.9
IC	47	15.8	34.8	20.1	918	4.6
SP	24	10.4	17.8	17.7	504	2.5
All	300	110.2	222.2	19.2	6509	32.7

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 111 plots, 300 ft grid (orange: QA flag)

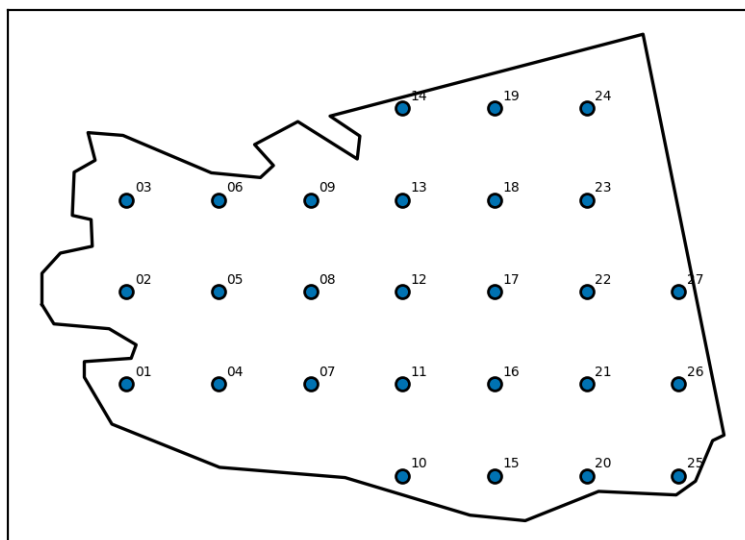


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
111-01	13	260	26	ok
111-02	11	220	14	ok
111-03	11	220	21	ok
111-04	9	180	21	ok
111-05	14	280	8	ok
111-06	10	200	14	ok
111-07	13	260	8	ok
111-08	9	180	7	ok
111-09	11	220	24	ok
111-10	8	160	29	ok
111-11	13	260	12	ok
111-12	14	280	13	ok
111-13	12	240	24	ok
111-14	10	200	32	ok
111-15	9	180	11	ok
111-16	11	220	17	ok
111-17	13	260	10	ok
111-18	13	260	15	ok
111-19	10	200	25	ok
111-20	9	180	27	ok
111-21	10	200	12	ok
111-22	12	240	17	ok
111-23	12	240	14	ok
111-24	12	240	18	ok
111-25	6	120	30	ok
111-26	11	220	18	ok
111-27	14	280	26	ok

Unit 112 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 80.5 ac. 40 plots, BAF 20, variable-radius cruise on a 300 ft grid, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 2.5 %)	Type	Sierran mixed conifer
Trees per acre	92	CWHR size / density	4 / D (cover 87 %)
QMD	20.6 in	SDI	279 (48 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	210 sq ft/ac 21.5 in	BA QMD	4 sq ft/ac 8.6 in
TPA	84	TPA	9
Volume	33.2 MBF/ac (6608 cu ft)	Volume	1.4 green tons/ac (64 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 156 sq ft/ac BA	Vol. sampling error	9.3 % at 95 % (CV 29 %, t = 2.02, n = 40)
Species (BA)	WF 66, PP 63, IC 36, DF 30, SP 19	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	4.1 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	5.1 % at 95 % (CV 16 %); reference, not the standard
		Measured trees	PP 126, WF 131, DF 61, SP 38, IC 72; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	40; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	2	3.4	1.0	15	0.1
8-11.9	19	16.0	9.5	206	1.0
12-15.9	47	21.9	23.5	523	2.6
16-19.9	62	17.6	31.0	787	3.9
20-23.9	61	11.7	30.5	865	4.3
24-27.9	69	9.5	34.5	1112	5.6
28-31.9	51	5.2	25.5	869	4.4
32-35.9	42	3.4	21.0	735	3.7
36-39.9	28	1.8	14.0	566	2.8
40-43.9	12	0.6	6.0	235	1.2
44-47.9	12	0.5	6.0	246	1.2
48-51.9	9	0.3	4.5	199	1.0
52-55.9	9	0.3	4.5	202	1.0
56-59.9	1	0.0	0.5	23	0.1
60-63.9	4	0.1	2.0	88	0.4
All	428	92.5	214.0	6672	33.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	131	27.0	65.5	21.1	2068	10.4
PP	126	27.2	63.0	20.6	2148	10.8
IC	72	15.3	36.0	20.7	966	4.8
DF	61	14.7	30.5	19.5	904	4.5
SP	38	8.3	19.0	20.5	585	2.9
All	428	92.5	214.0	20.6	6672	33.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112 plots, 300 ft grid (orange: QA flag)

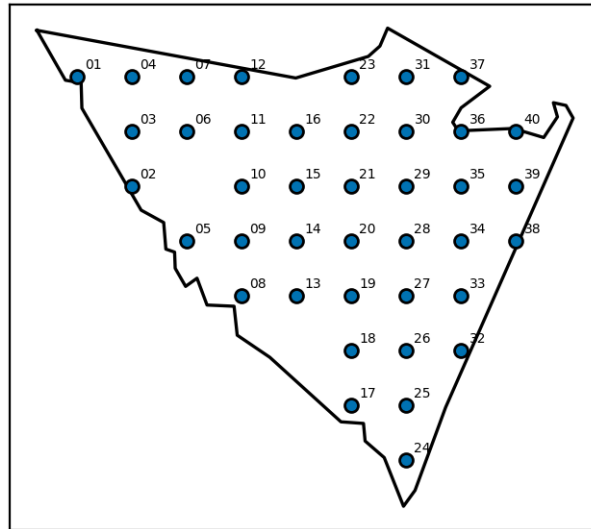


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
112-01	11	220	12	ok
112-02	11	220	10	ok
112-03	11	220	17	ok
112-04	12	240	12	ok
112-05	9	180	21	ok
112-06	12	240	36	ok
112-07	10	200	10	ok
112-08	5	100	21	ok
112-09	12	240	10	ok
112-10	11	220	10	ok
112-11	9	180	10	ok
112-12	11	220	29	ok
112-13	10	200	12	ok
112-14	13	260	8	ok
112-15	13	260	15	ok
112-16	13	260	16	ok
112-17	10	200	13	ok
112-18	12	240	36	ok
112-19	9	180	13	ok
112-20	11	220	21	ok
112-21	10	200	14	ok
112-22	13	260	13	ok
112-23	10	200	22	ok
112-24	13	260	28	ok
112-25	10	200	26	ok
112-26	12	240	11	ok
112-27	8	160	9	ok
112-28	10	200	15	ok
112-29	8	160	13	ok
112-30	11	220	12	ok
112-31	8	160	7	ok
112-32	12	240	25	ok
112-33	11	220	26	ok
112-34	11	220	20	ok
112-35	10	200	10	ok
112-36	13	260	39	ok
112-37	11	220	16	ok
112-38	9	180	16	ok
112-39	12	240	19	ok
112-40	11	220	30	ok

Unit 113 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 73.2 ac. 36 plots, BAF 20, variable-radius cruise on a 300 ft grid, 411 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	228 sq ft/ac (1 SE of mean, BA 2.0 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 95 %)
QMD	24.5 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	228 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	70	TPA	0
Volume	39.7 MBF/ac (7904 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 162 sq ft/ac BA	Vol. sampling error	6.1 % at 95 % (CV 18 %, t = 2.03, n = 36)
Species (BA)	PP 70, WF 66, IC 36, DF 34, SP 23	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	5.6 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.1 % at 95 % (CV 12 %); reference, not the standard
		Measured trees	PP 126, WF 118, DF 62, SP 41, IC 64; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	36; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	2.7	1.7	42	0.2
12-15.9	18	9.2	10.0	259	1.3
16-19.9	45	13.9	25.0	676	3.4
20-23.9	81	17.3	45.0	1363	6.8
24-27.9	70	10.6	38.9	1290	6.5
28-31.9	50	5.8	27.8	943	4.7
32-35.9	49	4.3	27.2	1039	5.2
36-39.9	43	3.1	23.9	999	5.0
40-43.9	21	1.2	11.7	500	2.5
44-47.9	13	0.6	7.2	311	1.6
48-51.9	5	0.2	2.8	127	0.6
52-55.9	5	0.2	2.8	127	0.6
56-59.9	4	0.1	2.2	107	0.5
60-63.9	4	0.1	2.2	120	0.6
All	411	69.5	228.3	7904	39.7

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	126	20.3	70.0	25.2	2704	13.6
WF	118	19.6	65.6	24.8	2300	11.5
IC	64	10.7	35.6	24.6	999	5.0
DF	62	12.7	34.4	22.3	1093	5.5
SP	41	6.2	22.8	25.9	808	4.1
All	411	69.5	228.3	24.5	7904	39.7

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 113 plots, 300 ft grid (orange: QA flag)

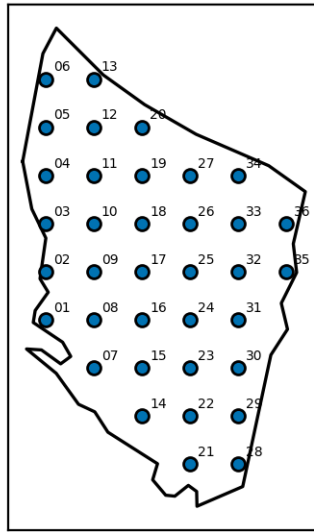


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
113-01	13	260	26	ok
113-02	14	280	27	ok
113-03	11	220	27	ok
113-04	11	220	24	ok
113-05	13	260	28	ok
113-06	13	260	31	ok
113-07	9	180	21	ok
113-08	13	260	16	ok
113-09	11	220	35	ok
113-10	12	240	21	ok
113-11	12	240	33	ok
113-12	11	220	20	ok
113-13	11	220	23	ok
113-14	12	240	23	ok
113-15	11	220	33	ok
113-16	13	260	25	ok
113-17	11	220	22	ok
113-18	12	240	11	ok
113-19	9	180	32	ok
113-20	10	200	19	ok
113-21	12	240	35	ok
113-22	10	200	30	ok
113-23	12	240	28	ok
113-24	12	240	27	ok
113-25	10	200	25	ok
113-26	10	200	27	ok
113-27	10	200	32	ok
113-28	10	200	21	ok
113-29	12	240	17	ok
113-30	12	240	10	ok
113-31	9	180	10	ok
113-32	11	220	9	ok
113-33	12	240	15	ok
113-34	15	300	17	ok
113-35	11	220	13	ok
113-36	11	220	11	ok

Unit 405 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 42.6 ac. 20 plots, BAF 20, variable-radius cruise on a 300 ft grid, 236 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	236 sq ft/ac (1 SE of mean, BA 2.3 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 95 %)
QMD	26.6 in	SDI	278 (47 % of max 593)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	235 sq ft/ac 27.0 in	BA QMD	1 sq ft/ac 9.5 in
TPA	59	TPA	2
Volume	47.1 MBF/ac (9389 cu ft)	Volume	0.4 green tons/ac (16 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 176 sq ft/ac BA	Vol. sampling error	12.6 % at 95 % (CV 27 %, t = 2.09, n = 20)
Species (BA)	WF 75, PP 57, IC 38, DF 37, SP 29	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	3.7 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.7 % at 95 % (CV 10 %); reference, not the standard
		Measured trees	PP 57, WF 75, DF 37, SP 29, IC 38; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	20; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	5.3	3.0	59	0.3
12-15.9	10	9.4	10.0	234	1.2
16-19.9	20	11.0	20.0	558	2.8
20-23.9	22	8.6	22.0	710	3.6
24-27.9	34	9.3	34.0	1211	6.1
28-31.9	16	3.3	16.0	560	2.8
32-35.9	34	5.4	34.0	1320	6.6
36-39.9	23	3.0	23.0	1039	5.2
40-43.9	22	2.3	22.0	1059	5.3
44-47.9	14	1.2	14.0	689	3.5
48-51.9	9	0.7	9.0	426	2.1
52-55.9	12	0.8	12.0	583	2.9
56-59.9	6	0.3	6.0	355	1.8
60-63.9	11	0.6	11.0	604	3.0
All	236	61.0	236.0	9406	47.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	75	17.0	75.0	28.4	3005	15.1
PP	57	15.4	57.0	26.0	2603	13.1
IC	38	11.6	38.0	24.5	1262	6.3
DF	37	10.1	37.0	25.9	1326	6.7
SP	29	6.9	29.0	27.7	1210	6.1
All	236	61.0	236.0	26.6	9406	47.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 405 plots, 300 ft grid (orange: QA flag)

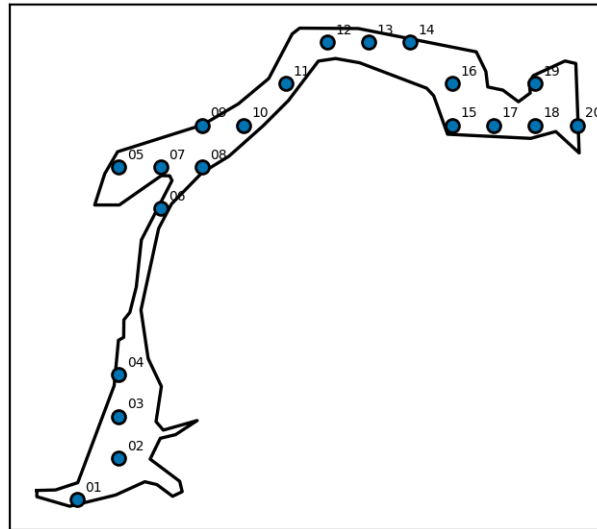


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
405-01	12	240	55	ok
405-02	12	240	70	ok
405-03	10	200	66	ok
405-04	12	240	59	ok
405-05	11	220	61	ok
405-06	11	220	48	ok
405-07	12	240	58	ok
405-08	12	240	48	ok
405-09	11	220	36	ok
405-10	12	240	65	ok
405-11	12	240	60	ok
405-12	13	260	60	ok
405-13	12	240	56	ok
405-14	15	300	37	ok
405-15	10	200	44	ok
405-16	14	280	49	ok
405-17	11	220	49	ok
405-18	11	220	53	ok
405-19	12	240	37	ok
405-20	11	220	50	ok

Unit 114 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 116.3 ac. 56 plots, BAF 20, variable-radius cruise on a 300 ft grid, 593 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (1 SE of mean, BA 2.1 %)	Type	Sierran mixed conifer
Trees per acre	93	CWHR size / density	4 / D (cover 85 %)
QMD	20.5 in	SDI	276 (48 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	208 sq ft/ac 21.6 in	BA QMD	4 sq ft/ac 7.9 in
TPA	81	TPA	11
Volume	32.5 MBF/ac (6468 cu ft)	Volume	1.4 green tons/ac (64 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 154 sq ft/ac BA	Vol. sampling error	7.7 % at 95 % (CV 29 %, t = 2.00, n = 56)
Species (BA)	PP 65, WF 61, IC 33, DF 30, SP 22	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	6.2 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.3 % at 95 % (CV 16 %); reference, not the standard
		Measured trees	PP 183, WF 172, DF 84, SP 62, IC 92; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	56; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	7.0	1.8	29	0.1
8-11.9	23	13.4	8.2	143	0.7
12-15.9	70	22.5	25.0	556	2.8
16-19.9	78	15.8	27.9	704	3.5
20-23.9	103	14.3	36.8	1024	5.1
24-27.9	70	7.0	25.0	765	3.8
28-31.9	79	5.7	28.2	967	4.9
32-35.9	59	3.4	21.1	768	3.9
36-39.9	30	1.3	10.7	409	2.1
40-43.9	25	0.9	8.9	359	1.8
44-47.9	21	0.7	7.5	324	1.6
48-51.9	9	0.2	3.2	141	0.7
52-55.9	5	0.1	1.8	79	0.4
56-59.9	7	0.1	2.5	109	0.5
60-63.9	9	0.2	3.2	155	0.8
All	593	92.8	211.8	6531	32.8

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	183	32.0	65.4	19.4	2158	10.8
WF	172	25.3	61.4	21.1	1945	9.8
IC	92	13.8	32.9	20.9	845	4.2
DF	84	12.6	30.0	20.9	904	4.5
SP	62	9.1	22.1	21.1	681	3.4
All	593	92.8	211.8	20.5	6531	32.8

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 114 plots, 300 ft grid (orange: QA flag)

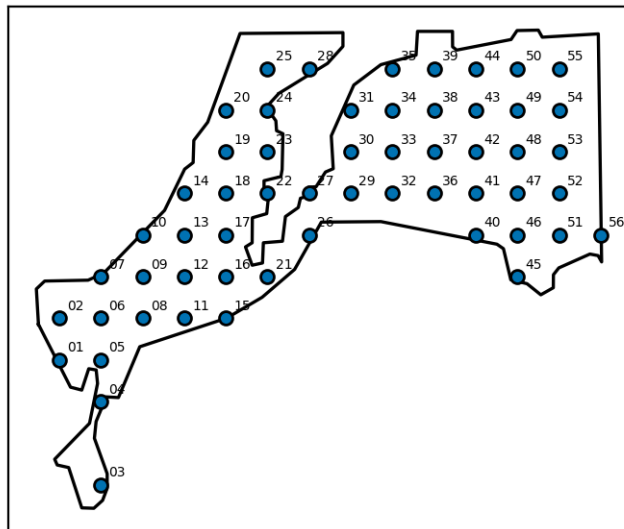


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-01	12	240	28	ok
114-02	9	180	17	ok
114-03	8	160	13	ok
114-04	12	240	21	ok
114-05	9	180	22	ok
114-06	11	220	11	ok
114-07	10	200	31	ok
114-08	14	280	21	ok
114-09	11	220	8	ok
114-10	12	240	32	ok
114-11	12	240	17	ok
114-12	12	240	21	ok
114-13	9	180	10	ok
114-14	11	220	19	ok
114-15	10	200	36	ok
114-16	11	220	11	ok
114-17	11	220	8	ok
114-18	10	200	20	ok
114-19	13	260	19	ok
114-20	12	240	18	ok
114-21	12	240	8	ok
114-22	9	180	9	ok
114-23	12	240	8	ok
114-24	8	160	10	ok
114-25	12	240	16	ok
114-26	9	180	28	ok
114-27	9	180	13	ok
114-28	11	220	21	ok
114-29	8	160	8	ok
114-30	12	240	14	ok
114-31	10	200	9	ok
114-32	10	200	10	ok
114-33	12	240	9	ok
114-34	11	220	13	ok
114-35	8	160	10	ok
114-36	6	120	10	ok
114-37	13	260	9	ok
114-38	12	240	43	ok
114-39	11	220	9	ok
114-40	11	220	23	ok
114-41	10	200	10	ok
114-42	11	220	9	ok
114-43	10	200	14	ok
114-44	10	200	11	ok
114-45	14	280	30	ok
114-46	10	200	18	ok
114-47	12	240	10	ok
114-48	8	160	9	ok
114-49	7	140	10	ok
114-50	10	200	9	ok
114-51	10	200	10	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-52	10	200	8	ok
114-53	12	240	6	ok
114-54	10	200	11	ok
114-55	11	220	8	ok
114-56	13	260	8	ok

SIMULATED DATA

Unit 115 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 84.0 ac. 39 plots, BAF 20, variable-radius cruise on a 300 ft grid, 444 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	228 sq ft/ac (1 SE of mean, BA 2.0 %)	Type	Sierran mixed conifer
Trees per acre	87	CWHR size / density	4 / D (cover 92 %)
QMD	21.9 in	SDI	292 (51 % of max 578)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	227 sq ft/ac 22.2 in	BA QMD	1 sq ft/ac 8.2 in
TPA	85	TPA	3
Volume	35.7 MBF/ac (7114 cu ft)	Volume	0.4 green tons/ac (18 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 158 sq ft/ac BA	Vol. sampling error	8.1 % at 95 % (CV 25 %, t = 2.02, n = 39)
Species (BA)	WF 68, PP 67, IC 34, DF 33, SP 26	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 3
		Sale volume share	5.5 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.0 % at 95 % (CV 12 %); reference, not the standard
		Measured trees	PP 130, WF 132, DF 65, SP 51, IC 66; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	39; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	1.8	0.5	9	0.0
8-11.9	11	8.8	5.6	114	0.6
12-15.9	43	20.2	22.1	513	2.6
16-19.9	64	18.7	32.8	794	4.0
20-23.9	73	14.2	37.4	1068	5.4
24-27.9	68	9.8	34.9	1110	5.6
28-31.9	51	5.4	26.2	883	4.4
32-35.9	48	3.9	24.6	889	4.5
36-39.9	31	2.1	15.9	593	3.0
40-43.9	22	1.2	11.3	437	2.2
44-47.9	15	0.7	7.7	312	1.6
48-51.9	9	0.3	4.6	211	1.1
52-55.9	5	0.2	2.6	112	0.6
60-63.9	3	0.1	1.5	88	0.4
All	444	87.3	227.7	7132	35.8

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	132	24.4	67.7	22.6	2139	10.7
PP	130	24.9	66.7	22.2	2313	11.6
IC	66	14.5	33.8	20.7	839	4.2
DF	65	12.4	33.3	22.2	1045	5.2
SP	51	11.2	26.2	20.7	795	4.0
All	444	87.3	227.7	21.9	7132	35.8

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 115 plots, 300 ft grid (orange: QA flag)

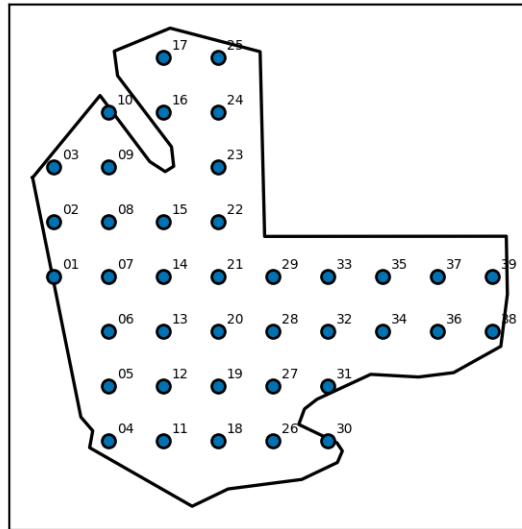


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
115-01	13	260	21	ok
115-02	10	200	12	ok
115-03	8	160	10	ok
115-04	12	240	20	ok
115-05	12	240	26	ok
115-06	12	240	17	ok
115-07	11	220	24	ok
115-08	11	220	13	ok
115-09	11	220	16	ok
115-10	9	180	30	ok
115-11	12	240	4	ok
115-12	12	240	12	ok
115-13	10	200	15	ok
115-14	10	200	10	ok
115-15	12	240	19	ok
115-16	9	180	24	ok
115-17	11	220	27	ok
115-18	11	220	9	ok
115-19	8	160	10	ok
115-20	11	220	7	ok
115-21	11	220	17	ok
115-22	14	280	16	ok
115-23	12	240	18	ok
115-24	12	240	15	ok
115-25	12	240	15	ok
115-26	10	200	6	ok
115-27	12	240	8	ok
115-28	12	240	13	ok
115-29	12	240	8	ok
115-30	11	220	17	ok
115-31	12	240	14	ok
115-32	12	240	5	ok
115-33	12	240	7	ok
115-34	12	240	5	ok
115-35	13	260	6	ok
115-36	15	300	3	ok
115-37	11	220	4	ok
115-38	12	240	15	ok
115-39	12	240	4	ok

Unit 116 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 89.2 ac. 44 plots, BAF 20, variable-radius cruise on a 300 ft grid, 507 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	230 sq ft/ac (1 SE of mean, BA 1.5 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 94 %)
QMD	24.6 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	230 sq ft/ac 24.9 in	BA QMD	1 sq ft/ac 9.4 in
TPA	68	TPA	2
Volume	40.1 MBF/ac (7993 cu ft)	Volume	0.5 green tons/ac (19 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 163 sq ft/ac BA	Vol. sampling error	4.5 % at 95 % (CV 15 %, t = 2.02, n = 44)
Species (BA)	PP 65, WF 62, DF 42, IC 36, SP 25	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 1
		Sale volume share	6.7 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.1 % at 95 % (CV 10 %); reference, not the standard
		Measured trees	PP 144, WF 136, DF 93, SP 55, IC 79; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	44; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	4.9	2.7	54	0.3
12-15.9	27	10.5	12.3	294	1.5
16-19.9	57	14.7	25.9	689	3.5
20-23.9	71	12.3	32.3	946	4.7
24-27.9	86	10.7	39.1	1295	6.5
28-31.9	70	6.4	31.8	1147	5.8
32-35.9	58	4.3	26.4	1002	5.0
36-39.9	44	2.6	20.0	829	4.2
40-43.9	29	1.4	13.2	546	2.7
44-47.9	15	0.6	6.8	287	1.4
48-51.9	21	0.7	9.5	422	2.1
52-55.9	10	0.3	4.5	214	1.1
56-59.9	5	0.1	2.3	112	0.6
60-63.9	8	0.2	3.6	175	0.9
All	507	69.7	230.5	8012	40.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	144	17.6	65.5	26.1	2562	12.9
WF	136	19.6	61.8	24.0	2143	10.8
DF	93	13.1	42.3	24.4	1386	7.0
IC	79	11.8	35.9	23.6	1001	5.0
SP	55	7.6	25.0	24.6	919	4.6
All	507	69.7	230.5	24.6	8012	40.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 116 plots, 300 ft grid (orange: QA flag)

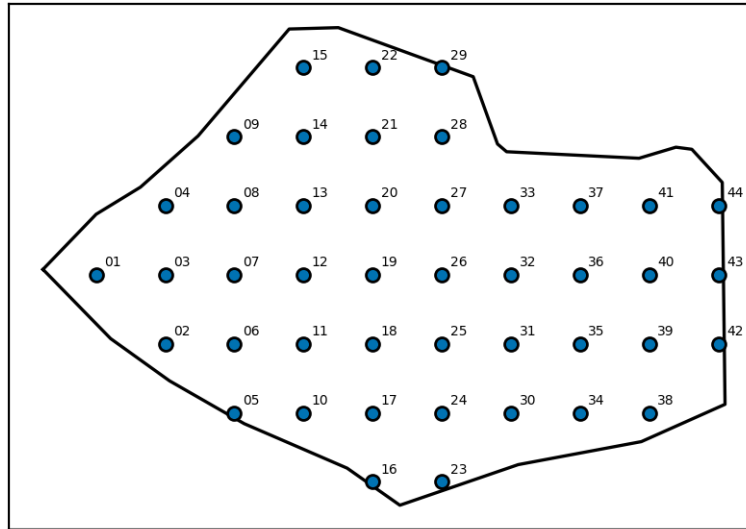


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
116-01	10	200	17	ok
116-02	12	240	17	ok
116-03	11	220	11	ok
116-04	10	200	20	ok
116-05	11	220	24	ok
116-06	12	240	15	ok
116-07	10	200	14	ok
116-08	10	200	11	ok
116-09	12	240	14	ok
116-10	11	220	19	ok
116-11	11	220	21	ok
116-12	14	280	12	ok
116-13	11	220	10	ok
116-14	11	220	4	ok
116-15	12	240	3	ok
116-16	10	200	15	ok
116-17	13	260	12	ok
116-18	11	220	16	ok
116-19	12	240	20	ok
116-20	12	240	12	ok
116-21	11	220	8	ok
116-22	10	200	4	ok
116-23	11	220	12	ok
116-24	13	260	12	ok
116-25	11	220	20	ok
116-26	11	220	14	ok
116-27	13	260	11	ok
116-28	13	260	16	ok
116-29	13	260	27	ok
116-30	11	220	22	ok
116-31	12	240	20	ok
116-32	14	280	12	ok
116-33	11	220	6	ok
116-34	10	200	10	ok
116-35	11	220	14	ok
116-36	11	220	11	ok
116-37	12	240	11	ok
116-38	13	260	13	ok
116-39	12	240	9	ok
116-40	13	260	10	ok
116-41	9	180	5	ok
116-42	11	220	11	ok
116-43	13	260	4	ok
116-44	12	240	19	ok

Unit 117 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 82.2 ac. 44 plots, BAF 20, variable-radius cruise on a 300 ft grid, 490 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	223 sq ft/ac (1 SE of mean, BA 2.0 %)	Type	Sierran mixed conifer
Trees per acre	89	CWHR size / density	4 / D (cover 88 %)
QMD	21.4 in	SDI	286 (49 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	219 sq ft/ac 22.4 in	BA QMD	4 sq ft/ac 8.7 in
TPA	80	TPA	9
Volume	35.0 MBF/ac (6963 cu ft)	Volume	1.3 green tons/ac (59 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 158 sq ft/ac BA	Vol. sampling error	7.7 % at 95 % (CV 25 %, t = 2.02, n = 44)
Species (BA)	WF 69, PP 66, IC 36, DF 33, SP 18	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 3
		Sale volume share	4.7 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.0 % at 95 % (CV 13 %); reference, not the standard
		Measured trees	PP 146, WF 151, DF 73, SP 40, IC 80; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	44; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	1.4	0.5	6	0.0
8-11.9	20	16.0	9.1	166	0.8
12-15.9	40	17.3	18.2	392	2.0
16-19.9	67	17.1	30.5	763	3.8
20-23.9	83	14.2	37.7	1077	5.4
24-27.9	71	8.9	32.3	1044	5.2
28-31.9	66	6.2	30.0	1036	5.2
32-35.9	50	3.6	22.7	811	4.1
36-39.9	31	1.8	14.1	544	2.7
40-43.9	21	1.0	9.5	397	2.0
44-47.9	19	0.7	8.6	371	1.9
48-51.9	9	0.3	4.1	168	0.8
52-55.9	7	0.2	3.2	137	0.7
56-59.9	3	0.1	1.4	68	0.3
60-63.9	2	0.0	0.9	42	0.2
All	490	89.0	222.7	7022	35.3

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	151	26.6	68.6	21.8	2191	11.0
PP	146	28.4	66.4	20.7	2289	11.5
IC	80	15.3	36.4	20.8	965	4.8
DF	73	12.5	33.2	22.0	986	4.9
SP	40	6.1	18.2	23.3	591	3.0
All	490	89.0	222.7	21.4	7022	35.3

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 117 plots, 300 ft grid (orange: QA flag)

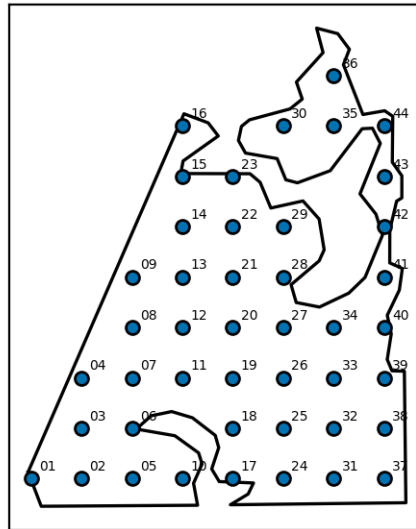


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
117-01	11	220	23	ok
117-02	12	240	37	ok
117-03	13	260	28	ok
117-04	13	260	9	ok
117-05	10	200	22	ok
117-06	10	200	39	ok
117-07	11	220	23	ok
117-08	12	240	30	ok
117-09	9	180	22	ok
117-10	11	220	21	ok
117-11	12	240	14	ok
117-12	11	220	15	ok
117-13	13	260	12	ok
117-14	12	240	18	ok
117-15	12	240	35	ok
117-16	10	200	27	ok
117-17	9	180	30	ok
117-18	11	220	20	ok
117-19	11	220	25	ok
117-20	12	240	25	ok
117-21	12	240	11	ok
117-22	13	260	9	ok
117-23	10	200	30	ok
117-24	11	220	34	ok
117-25	6	120	19	ok
117-26	11	220	27	ok
117-27	13	260	36	ok
117-28	12	240	22	ok
117-29	12	240	19	ok
117-30	13	260	25	ok
117-31	10	200	33	ok
117-32	10	200	21	ok
117-33	12	240	22	ok
117-34	11	220	10	ok
117-35	12	240	22	ok
117-36	11	220	26	ok
117-37	11	220	36	ok
117-38	13	260	17	ok
117-39	12	240	15	ok
117-40	12	240	33	ok
117-41	9	180	18	ok
117-42	9	180	33	ok
117-43	11	220	27	ok
117-44	9	180	28	ok

Unit 118 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 70.2 ac. 34 plots, BAF 20, variable-radius cruise on a 300 ft grid, 374 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	220 sq ft/ac (1 SE of mean, BA 3.2 %)	Type	Sierran mixed conifer
Trees per acre	76	CWHR size / density	4 / D (cover 91 %)
QMD	23.1 in	SDI	278 (47 % of max 595)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	219 sq ft/ac 23.4 in	BA QMD	1 sq ft/ac 8.7 in
TPA	73	TPA	3
Volume	36.0 MBF/ac (7180 cu ft)	Volume	0.6 green tons/ac (26 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 165 sq ft/ac BA	Vol. sampling error	7.1 % at 95 % (CV 20 %, t = 2.03, n = 34)
Species (BA)	WF 76, PP 59, DF 38, IC 35, SP 13	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	4.8 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	6.4 % at 95 % (CV 18 %); reference, not the standard
		Measured trees	PP 100, WF 129, DF 64, SP 22, IC 59; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	34; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	6.9	4.1	98	0.5
12-15.9	29	14.9	17.1	419	2.1
16-19.9	42	13.8	24.7	694	3.5
20-23.9	73	16.5	42.9	1233	6.2
24-27.9	67	10.7	39.4	1283	6.4
28-31.9	43	5.2	25.3	869	4.4
32-35.9	35	3.4	20.6	710	3.6
36-39.9	28	2.1	16.5	618	3.1
40-43.9	11	0.7	6.5	272	1.4
44-47.9	19	1.0	11.2	478	2.4
48-51.9	6	0.3	3.5	144	0.7
52-55.9	3	0.1	1.8	80	0.4
56-59.9	4	0.1	2.4	111	0.6
60-63.9	7	0.2	4.1	199	1.0
All	374	75.9	220.0	7206	36.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	129	29.5	75.9	21.7	2445	12.3
PP	100	21.6	58.8	22.4	2131	10.7
DF	64	11.7	37.6	24.3	1210	6.1
IC	59	9.4	34.7	26.0	978	4.9
SP	22	3.7	12.9	25.5	442	2.2
All	374	75.9	220.0	23.1	7206	36.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
118-01	12	240	17	ok
118-02	10	200	15	ok
118-03	12	240	12	ok
118-04	9	180	3	ok
118-05	15	300	16	ok
118-06	12	240	15	ok
118-07	11	220	13	ok
118-08	11	220	27	ok
118-09	10	200	17	ok
118-10	5	100	27	ok
118-11	10	200	22	ok
118-12	13	260	22	ok
118-13	12	240	18	ok
118-14	12	240	14	ok
118-15	11	220	15	ok
118-16	8	160	15	ok
118-17	12	240	17	ok
118-18	11	220	15	ok
118-19	11	220	8	ok
118-20	7	140	8	ok
118-21	13	260	15	ok
118-22	8	160	17	ok
118-23	11	220	15	ok
118-24	13	260	14	ok
118-25	10	200	9	ok
118-26	11	220	9	ok
118-27	10	200	5	ok
118-28	15	300	10	ok
118-29	13	260	9	ok
118-30	11	220	12	ok
118-31	12	240	9	ok
118-32	10	200	12	ok
118-33	12	240	18	ok
118-34	11	220	9	ok

Unit 406 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 35.1 ac. 17 plots, BAF 20, variable-radius cruise on a 300 ft grid, 205 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	241 sq ft/ac (1 SE of mean, BA 2.3 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	5 / D (cover 93 %)
QMD	24.9 in	SDI	297 (50 % of max 598)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	241 sq ft/ac 24.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	71	TPA	0
Volume	42.0 MBF/ac (8366 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 170 sq ft/ac BA	Vol. sampling error	9.5 % at 95 % (CV 18 %, t = 2.12, n = 17)
Species (BA)	WF 82, PP 60, IC 42, DF 34, SP 22	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	1.9 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.9 % at 95 % (CV 9 %); reference, not the standard
		Measured trees	PP 51, WF 70, DF 29, SP 19, IC 36; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	17; R5 20-plot minimum for an area-based cruise (sec. 41.3): SHORT by 3

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.6	1.2	24	0.1
12-15.9	12	12.2	14.1	358	1.8
16-19.9	17	11.5	20.0	577	2.9
20-23.9	41	18.1	48.2	1464	7.3
24-27.9	29	9.6	34.1	1113	5.6
28-31.9	34	8.2	40.0	1409	7.1
32-35.9	21	4.0	24.7	907	4.6
36-39.9	19	2.9	22.4	873	4.4
40-43.9	13	1.6	15.3	615	3.1
44-47.9	7	0.7	8.2	380	1.9
48-51.9	5	0.4	5.9	282	1.4
52-55.9	1	0.1	1.2	65	0.3
56-59.9	3	0.2	3.5	180	0.9
60-63.9	2	0.1	2.4	121	0.6
All	205	71.2	241.2	8366	42.0

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	70	21.1	82.4	26.7	3044	15.3
PP	51	21.5	60.0	22.6	2164	10.9
IC	36	12.1	42.4	25.4	1246	6.3
DF	29	10.2	34.1	24.8	1107	5.6
SP	19	6.3	22.4	25.5	806	4.0
All	205	71.2	241.2	24.9	8366	42.0

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 406 plots, 300 ft grid (orange: QA flag)

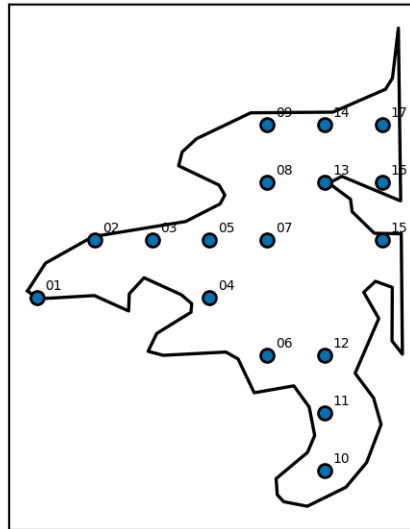


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted; 3 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
406-01	13	260	39	ok
406-02	12	240	45	ok
406-03	12	240	66	ok
406-04	13	260	40	ok
406-05	11	220	59	ok
406-06	14	280	35	ok
406-07	12	240	54	ok
406-08	11	220	67	ok
406-09	10	200	53	ok
406-10	11	220	61	ok
406-11	12	240	45	ok
406-12	11	220	35	ok
406-13	12	240	37	ok
406-14	13	260	48	ok
406-15	11	220	44	ok
406-16	13	260	41	ok
406-17	14	280	44	ok